Mahdi Alam

Toronto, ON | 647-916-4783 | mahdi.alam@torontomu.ca
mahdialam.com | github.com/mahdi-alam360 | linkedin.com/in/mahdi-alam-264127150

PROFILE

Computer Science graduate and MSc researcher with Python data processing, reproducible experimentation and SQL database projects. Built local HTTP APIs, parameterized SQL reports and a new AI-assisted CSV-to-SQLite ETL learning project with automated checks. Interested in early-career software, data and machine-learning roles.

EDUCATION

Toronto Metropolitan University | Toronto, ON

MSc, Computer Science | Thesis research in artificial intelligence
September 2023 - Expected May 2027 (tentative)

BSc (Honours), Computer Science | September 2019 - May 2023

TECHNICAL SKILLS

Programming: Python, C++, C; JavaScript, TypeScript and C# in personal/academic projects

Backend and databases: Node.js, JSON HTTP APIs, ASP.NET Core, SQLite, PHP and MySQL project experience; SQL queries and request validation

Tools and ML: Git, Docker, Linux, GitHub Actions (new learning project), PyTorch, scikit-learn, NumPy, Pandas

SELECTED SOFTWARE PROJECTS

Order Pipeline | Python, SQLite, CSV, SQL | October 2026

New AI-assisted learning project | mahdialam.com/order-pipeline/

- Built a local CSV-to-SQLite ETL pipeline with record validation, rejected-row reporting, integer-cent currency handling, transactional upserts and SQL aggregates. Six tests cover replay, updates, validation, exact totals and provenance. Synthetic data; no AWS deployment or professional ETL experience.

Document Search Lab | React, TypeScript, Node.js | October 2026

AI-assisted personal project | mahdialam.com/search-lab/

- Built a BM25 search interface/API with custom collections, per-term explanations and request validation; verified 15 automated ranking, input-validation and HTTP tests, type checks and build.

RoadFrame - Imagery Review Lab | C#, ASP.NET Core, SQLite | October 2026

AI-assisted learning project | mahdialam.com/roadframe/

- Built a local REST API with SQLite persistence, parameterized SQL, foreign keys, upserts and JOIN/GROUP BY/AVG reports. Verified 18 HTTP/SQL assertions including invalid input, updates, injection-shaped notes and persistence after restart.

Model Lab | Python, JavaScript, GitHub Actions | October 2026

New AI-assisted learning project | mahdialam.com/model-lab/

- Exported deterministic logistic-model parameters, provenance and held-out synthetic evaluation to a validated serverless prediction API. Verified 15 training/API tests locally and in GitHub Actions, plus live inference on Vercel. Synthetic demonstration; not professional MLOps experience.

RESEARCH EXPERIENCE

MSc Thesis Research - Self-Supervised Medical Image Registration
Toronto Metropolitan University | Toronto, ON

- Designed a two-phase, coarse-to-fine training strategy combining multi-resolution curriculum learning and adaptive optimization for self-supervised deep learning.
- Improved Dice Similarity Coefficient by up to 1.7 points across public medical-image datasets (LPBA40, IXI, OASIS), using quantitative benchmark evaluation.
- Evaluated deformation quality through Jacobian determinant analysis and conducted reproducible Python experiments with ablation studies.

AVAILABILITY

Canadian citizen; authorized to work without sponsorship. Toronto-based; available for full-time office work in Toronto. Available one month after an offer. MSc expected May 2027 (tentative). Research manuscript in preparation.